

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE		
TASK:	Perform a Manual AFD Calculation		
TASK NUMBER:	N0150010201		
JPM NUMBER:	11-01 NRC RO Admin A1-2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.18
APPLICABILITY:	IMPORTANCE FACTOR: <u>3.6</u>		
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S1.OP-ST.NIS-0001, Rev. 9, Power Distribution – Axial Flux Difference Salem Unit 1 Core Operating Limits Report Cycle 22, Rev. 4 S1.RE-RA.ZZ-0011(Q) Tables, Rev. 343		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	<u>15 min</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	11-05-12
Validated By:	D Norgaard SME or Instructor	Date:	11-8-12
Approved By:	<i>[Signature]</i> (Blos E) Training Department	Date:	11/9/12
Approved By:	<i>[Signature]</i> (Brynn L. L. L.) Operations Department	Date:	11/9/12
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Administrative

TASK: Perform a Manual AFD Calculation

TASK NUMBER: N0150010201

INITIAL CONDITIONS: Salem Unit 1 completed a down power to 73% power 15 minutes ago IAW TSAS 3.1.3.1.c.3.d due to a single control rod being declared inoperable. The AFD Monitor Alarm for the Auxiliary Annunciator was determined to be inoperable 10 minutes ago.

INITIATING CUE: You are directed to perform S1.OP-ST.NIS-0001(Q), Power Distribution – Axial Flux Difference in response to the AFD Monitor Alarm being inoperable.

Current AFD Channel Readings are:

Channel 1: - 8.9

Channel II: -8.6

Channel III: -10.2

Channel IV: - 9.3

The component Off Normal and Off Normal Tagged lists have been reviewed and no systems or support systems are affected for this procedure.

Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Completes applicable portion of Attachment 3 of S1.OP-ST.NIS-0001, and determines Channel III is outside the AFD Target Band.

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NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Perform a Manual AFD Calculation

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide blank copies of S1.OP-ST.NIS-0001, Power Distribution – Axial Flux Difference, a copy of Salem Unit 1 Core Operating Limits Report, and S1.RE-RA.ZZ-0011 Tables Rev. 343			
	2.1	IDENTIFY sections of this procedure that are <u>NOT</u> to be performed with “N/A”.	Identifies Sections 5.1 and 5.2 will not be performed.		
	2.2	REVIEW components “Off Normal and Off Normal Tagged” List(s) for the system and support system(s) associated with the evolution to be performed in this procedure.	Determines from initial conditions that the component Off Normal and Off Normal Tagged lists have been reviewed and no systems or support systems are affected for this procedure.		
	2.3	Unit is in Mode 1 <u>AND</u> >15% RTP.	Determines from initial conditions that the Unit is in Mode 1 >15% power.		
	3.0	Precautions and Limitations	Reviews and initials Precautions and Limitations.		

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System: ADMINISTRATIVE

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.3.1	IF AFD Monitor (XA-8740, OR Monitor Alarm (Auxiliary Annunciator)) is inoperable, THEN COMPLETE Attachment 3, AFD Data Log Sheet as follows:			
	5.3.1.A	ENTER present Reactor Power	Enters 73% power		
	5.3.1.B	ENTER AFD Target from S1.RE-RA.ZZ-0011(Q) Tables, Table 3.	Enters -1.5 for AFD Target (100% Value)		
*	5.3.1.C	CALCULATE the AFD Target Value IAW the following: $A \times B = \text{AFD Target Value (present power)} = C$ 100% Where:(A)= Present Reactor Power (B)= AFD Target (100% Value) (C)= AFD Target (present power)	Calculates AFD Target for present power level as: $\frac{73 \times -1.5}{100} = \mathbf{1.1}$		
*	5.3.1.D	DETERMINE AFD Target Band Upper and Lower Limits as follows: ___ +(C)= Upper Limit ___ -(C)= Lower Limit	Determines Upper Limit is: $+6 + (-1.1) = \mathbf{+4.9}$ Determines Lower Limit is: $-9 + (-1.1) = \mathbf{-10.1}$		
	5.3.1.E	ENTER AFD Channel Readings.	Enters AFD values from initial conditions		

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.3.1.F	DETERMINE if any Channel is operating outside of the Target Band Upper or Lower Limits (inclusive)	Determines Channel III (-10.2) is operating outside the Lower Target Band.		
	5.3.1.G	REQUEST a second operator to perform an independent verification of the calculations performed on Attachment 3.	Cue: IV of calculations performed on Attachment 3 is complete SAT.		
	5.3.1.H	<u>IF</u> all AFD Channels are within the AFD Target Band, <u>THEN CONTINUE</u> to record data once per hour for the first 24 hours of inoperability, and every 30 minutes thereafter, until restored to OPERABLE status.	Determines that one AFD Channel is NOT within the Target Band.		
	5.3.1.I	<u>IF</u> one or more AFD Channels indicate outside the AFD Target Band, <u>THEN CONTINUE</u> to record data on Attachment 3 IAW Section 5.4, AFD Alarm Condition.	Determines continued readings are required IAW Section 5.4		
			When determination of continued data recording requirement by using Section 5.4 is made, terminate JPM.		

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JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- S 1. Task description and number, JPM description and number are identified.
- S 2. Knowledge and Abilities (K/A) references are included.
- S 3. Performance location specified. (in-plant, control room, or simulator)
- S 4. Initial setup conditions are identified.
- S 5. Initiating and terminating Cues are properly identified.
- S 6. Task standards identified and verified by SME review.
- S 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- S 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev 11/25 Date 11-9-12
- S 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- N/D 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: Staudin Sr D. Norgaard

Date: 11-9-12

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

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